number of degrees of freedom. It's even true for the universe as a whole; because there is vacuum energy, the acceleration of space is expanding, and that means there is a horizon all around us that delineates the extent of the observable part of our cosmos. That observable patch of space has a finite maximum entropy, so there are only a finite number of degrees of freedom needed to describe everything we see or ever will see.

If this story is on the right track, it has an immediate, profound consequence for the Many-Worlds picture of quantum mechanics. A finite number of quantum degrees of freedom implies a finite-dimensional Hilbert space for the system as a whole (in this case, any chosen region of space). That in turn implies that there is some finite number of branches of the wave function, not an infinite number. That's why Alice was cagey back in Chapter Eight about whether there are an infinite number of "worlds" in the wave function. In many simple models of quantum mechanics, including that of a fixed set of particles moving smoothly through space or any ordinary quantum field theory, Hilbert space is infinite-dimensional and there could potentially be an infinite number of worlds. But gravity seems to change things around in an important way. It prevents most of those worlds from existing, because they would describe too much energy being packed into a local region.

So maybe in the real universe, where gravity certainly exists, Everettian quantum mechanics only describes a finite number of worlds. The number Alice mentioned for the dimensionality of Hilbert space was $2^{10^{122}}$.

Now we can reveal where that number came from: it's from calculating the entropy that our observable universe will have once it reaches maximum entropy, and working backward to find out how big Hilbert space needs to be to accommodate that much entropy. (The size of the observable universe is set by the vacuum energy, so the exponent 10^{122} is the ratio of the Planck scale to the cosmological constant, familiar from our discussion in Chapter Twelve.) Our confidence in the basic principles of quantum gravity isn't strong enough to be absolutely sure that there are only a finite number of Everettian worlds, but it seems reasonable, and it certainly would make things much simpler.